STD 1: Measurement (Std 1)

M3 Right-Angled Triangles (Y12)

Teacher: Denis Vlismas

Exam Equivalent Time: 103.5 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*

IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- *MS-M3 Right-Angled Triangles* has accounted for 5% and 6% of the *2020 and 2019 Std1 exams* respectively.
- Although this specific topic has been a minor contributor to past Gen2 papers, the absence of non-right angled trigonometry in the Std1 syllabus create the likelihood of increased allocations, in our view.

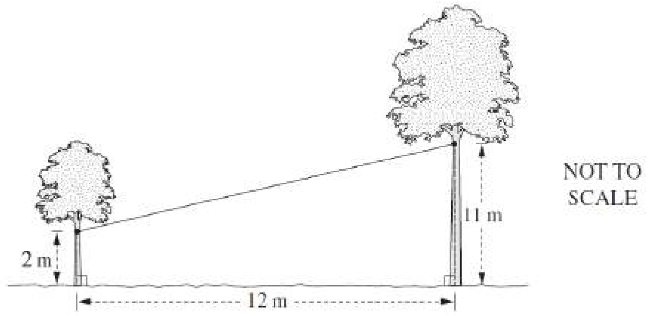
ANALYSIS - Common pitfalls

- *Right-Angled Triangles* attracted a 4-mark question on basic trig in the *2020 Std1 exam* which followed 2 separate questions in the *2019 Std1 exam* worth a total of 5-marks.
- In Gen2 exam history, this topic has been most commonly tested via multiple choice questions which are well represented in the database.
- Students must be confident in applying all trig ratios (see *2020 Std1 11*), compass and true bearings and be well versed in common terminology such as *angle of depression* and *angle of elevation*.
- This sub-topic provides a good opportunity to score well. The higher level of difficulty in this area can be seen in examples *2015 Std2 9 MC*, *2012 Std2 27d* and *2009 Std2 23a*.
- Rounding an angle *to the nearest minute* is specifically mentioned in the new syllabus and calculator proficiency in this area is important.
- In response to the above point, we have adjusted "nearest degree" in some past HSC questions to "nearest minute", testing students with some harder rounding examples.

Questions

1. Measurement, STD2 M6 2011 HSC 9 MC

Two trees on level ground, 12 metres apart, are joined by a cable. It is attached 2 metres above the ground to one tree and 11 metres above the ground to the other.



What is the length of the cable between the two trees, correct to the nearest metre?

(A) 9 m

(B) 12 m

(C) 15 m

(D) 16 m

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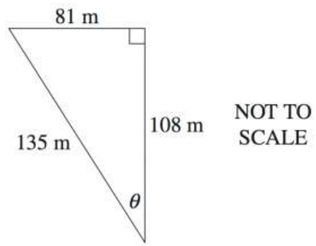
2. Measurement, STD2 M6 2013 HSC 4 MC

What is the value of θ, to the nearest degree?



(A) $31°$

(B) $37°$

(C) $49°$

(D) $53°$

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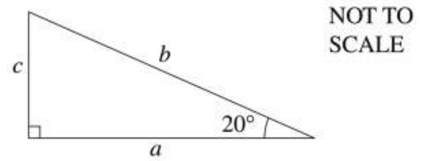
3. Measurement, STD2 M6 2004 HSC 5 MC

What is the correct expression for tan 20° in this triangle?



(A) $\dfrac{a}{b}$

(B) $\dfrac{a}{c}$

(C) $\dfrac{c}{b}$

(D) $\dfrac{c}{a}$

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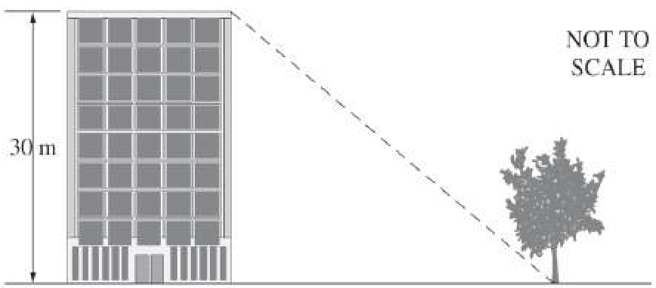
4. Measurement, STD2 M6 2006 HSC 3 MC

The angle of depression of the base of the tree from the top of the building is 65°. The height of the building is 30 m.

How far away is the base of the tree from the building, correct to one decimal place?



(A) 12.7 m

(B) 14.0 m

(C) 33.1 m

(D) 64.3 m

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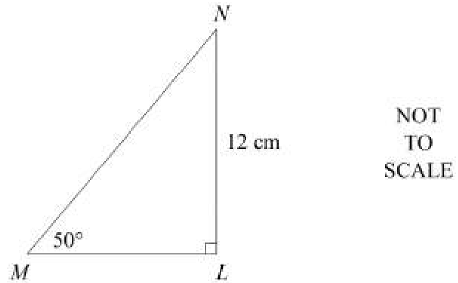
5. Measurement, STD2 M6 2007 HSC 8 MC

What is the length of the side MN in the following triangle, correct to two decimal places?



(A) 9.19 cm

(B) 10.07 cm

(C) 15.66 cm

(D) 18.67 cm

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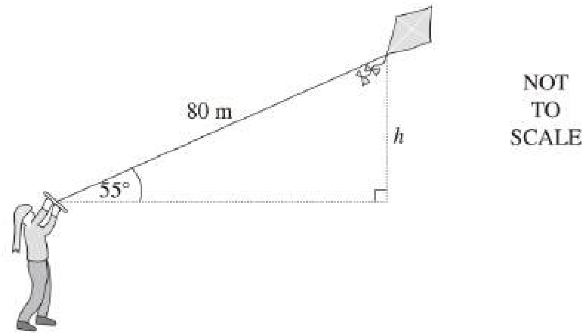
6. Measurement, STD2 M6 2008 HSC 14 MC

Danni is flying a kite that is attached to a string of length 80 metres. The string makes an angle of 55° with the horizontal.

How high, to the nearest metre, is the kite above Danni’s hand?



(A) 46 m

(B) 66 m

(C) 98 m

(D) 114 m

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7. Measurement, STD2 M6 2005 HSC 8 MC

If $\tan\theta = 85$, what is the value of θ, correct to 2 decimal places?

(A) 1.37°

(B) 1.56°

(C) 89.33°

(D) 89.20°

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8. Measurement, STD2 M1 2019 HSC 4 MC

Which compass bearing is the same as a true bearing of 110°?

A. S20°E

B. S20°W

C. S70°E

D. S70°W

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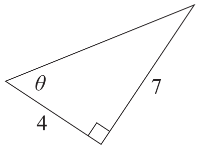
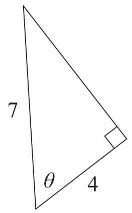
9. Measurement, STD2 M6 SM-Bank 1 MC

In which triangle is $\sin\theta = \dfrac{4}{7}$?

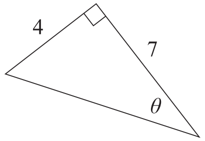
A.



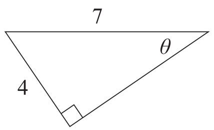
B.



C.



D.



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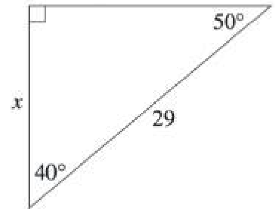
10. Measurement, STD2 M6 2012 HSC 4 MC

Which expression could be used to calculate the value of x in this triangle?



NOT TO SCALE

(A) $29 \times \cos 40^\circ$

(B) $29 \times \cos 50^\circ$

(C) $\dfrac{\cos 40^\circ}{29}$

(D) $\dfrac{\cos 50^\circ}{29}$

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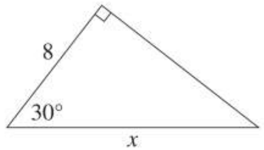
11. Measurement, STD2 M6 2009 HSC 4 MC

Which is the correct expression for the value of x in this triangle?



NOT TO SCALE

(A) $\dfrac{8}{\cos 30^\circ}$

(B) $\dfrac{8}{\sin 30^\circ}$

(C) $8 \times \cos 30^\circ$

(D) $8 \times \sin 30^\circ$

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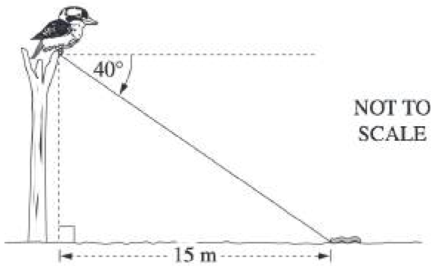
12. Measurement, STD2 M6 2011 HSC 4 MC

The angle of depression from a kookaburra's feet to a worm on the ground is 40°. The worm is 15 metres from a point on the ground directly below the kookaburra's feet.



How high above the ground are the kookaburra's feet, correct to the nearest metre?

(A) 10 m

(B) 11 m

(C) 13 m

(D) 18 m

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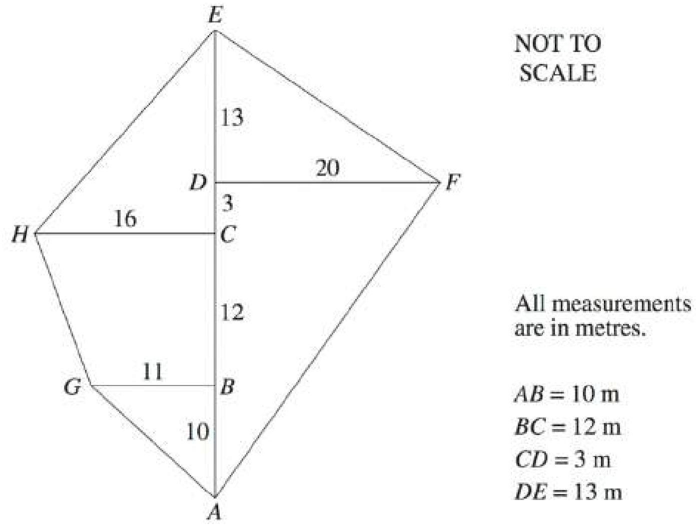
13. Measurement, STD2 M6 2010 HSC 3 MC

A field diagram has been drawn from an offset survey.



What is the distance from G to H correct to the nearest metre?

(A) 11

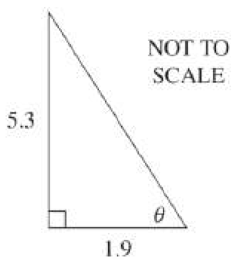
(B) 13

(C) 16

(D) 20

14. Measurement, STD2 M6 2017 HSC 8 MC

The diagram shows a right-angled triangle.



What is the value of θ, to the nearest minute?

A. $70^\circ 16'$

B. $70^\circ 17'$

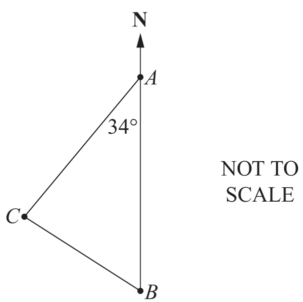
C. $70^\circ 27'$

D. $70^\circ 28'$

15. Measurement, STD2 M6 2018 HSC 7 MC

The diagram shows the positions of towns A, B and C.

Town A is due north of town B and $\angle CAB = 34^\circ$



What is the bearing of town C from town A?

A. 034°

B. 146°

C. 214°

D. 326°

16. Measurement, STD2 M6 SM-Bank 3 MC

Which of the following expresses S65°W as a true bearing?

A. 065°

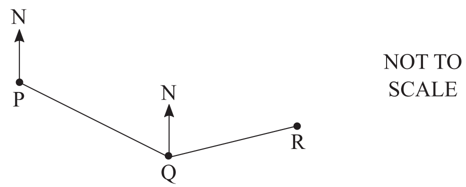
B. 155°

C. 245°

D. 295°

17. Measurement, STD2 M6 SM-Bank 4 MC

Ralph travels from P to Q on a bearing of 130°. He then turns and walks to R on a bearing of 075°.



What is the size of $\angle PQR$?

A. 95°

B. 100°

C. 115°

D. 125°

18. Measurement, STD2 M6 SM-Bank 7 MC

Jeet walks 5 km from his home on a bearing of 153°. He then walks due north until he arrives a point which is due east of his home.

How far east, to the nearest 0.1 km, is Jeet from home?

A. 2.3 km

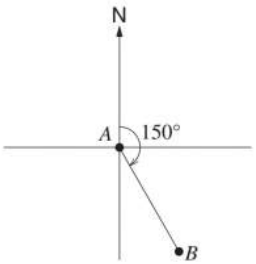
B. 2.5 km

C. 4.9 km

D. 9.8 km

19. Measurement, STD2 M6 2010 HSC 10 MC

A plane flies on a bearing of 150° from A to B.



NOT TO SCALE

What is the bearing of A from B?

(A) $30°$

(B) $150°$

(C) $210°$

(D) $330°$

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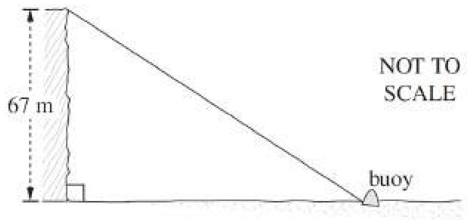
20. Measurement, STD2 M6 2015 HSC 9 MC

From the top of a cliff 67 metres above sea level, the angle of depression of a buoy is 42°.



NOT TO SCALE

How far is the buoy from the base of the cliff, to the nearest metre?

(A) 60 m

(B) 74 m

(C) 90 m

(D) 100 m

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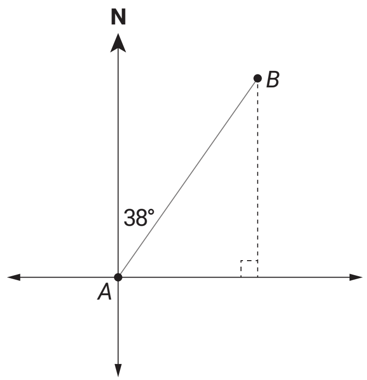
21. Measurement, STD1 M3 2021 HSC 10 MC

The compass bearing of B from A is N38°E.



NOT TO SCALE

What is the true bearing of A from B?

A. 128°

B. 218°

C. 232°

D. 322°

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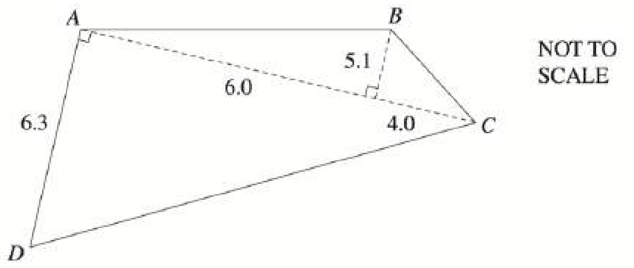
22. Measurement, STD2 M1 2004 HSC 23a

The diagram shows the shape of Carmel's garden bed. All measurements are in metres.



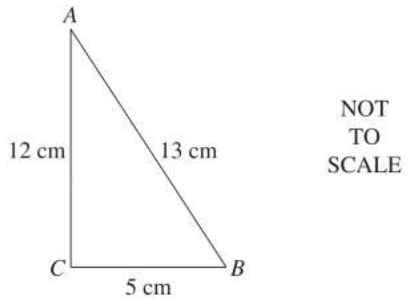
i. Show that the area of the garden bed is 57 square metres. ***(2 marks)***

ii. Carmel decides to add a 5 cm layer of straw to the garden bed.
Calculate the volume of straw required. Give your answer in cubic metres. ***(2 marks)***

iii. Each bag holds 0.25 cubic metres of straw.
How many bags does she need to buy? ***(2 marks)***

iv. A straight fence is to be constructed joining point A to point B.
Find the length of this fence to the nearest metre. ***(2 marks)***

23. Measurement, STD2 M6 2005 HSC 25b

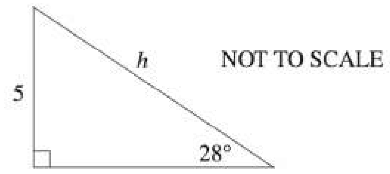


i. Use Pythagoras' theorem to show that ΔABC is a right-angled triangle. ***(1 mark)***

ii. Calculate the size of $\angle ABC$ to the nearest **minute**. ***(2 marks)***

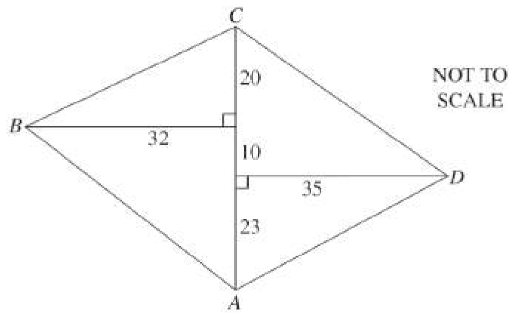
24. Measurement, STD2 M6 2014 HSC 26b

Calculate the value of h correct to two decimal places. ***(2 marks)***



25. Measurement, STD2 M6 2016 HSC 26d

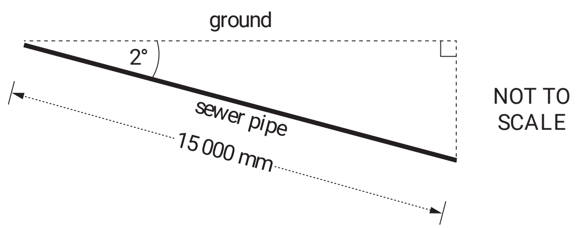
The diagram shows a block of land $ABCD$ that has been surveyed. All measurements are in metres.



Calculate the length of AB, correct to the nearest metre. ***(2 marks)***

26. Measurement, STD2 M6 2017 HSC 26d

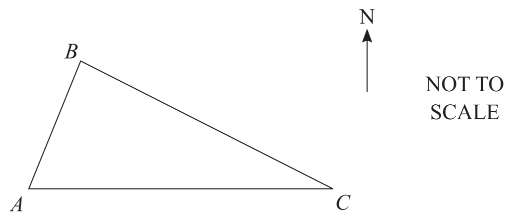
A sewer pipe needs to be placed into the ground so that it has a 2° angle of depression. The length of the pipe is 15 000 mm.



How much deeper should one end of the pipe be compared to the other end? Answer to the nearest mm. ***(2 marks)***

27. Measurement, STD2 M6 SM-Bank 4

The diagram shows three checkpoints *A*, *B* and *C*. Checkpoint *C* is due east of Checkpoint *A*. The bearing of Checkpoint *B* from Checkpoint *A* is N22°E and the bearing of Checkpoint *C* from Checkpoint *B* is S68°E. The distance between Checkpoint *A* and Checkpoint *B* is 42 kilometres.



i. Mark the given information on the diagram and explain why $\angle ABC$ is 90°. ***(2 marks)***

ii. Find the distance, to the nearest kilometre, between Checkpoint *A* and Checkpoint *C*. ***(2 marks)***

iii. If a runner is travelling 12.6 km/h, how long does it take her to travel between Checkpoint *A* and Checkpoint *B*, in hours and minutes? ***(2 marks)***

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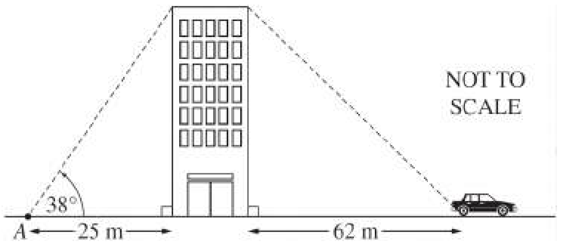
28. Measurement, STD2 M6 2009 HSC 23a

The point A is 25 m from the base of a building. The angle of elevation from A to the top of the building is 38°.



i. Show that the height of the building is approximately 19.5 m. ***(1 mark)***

ii. A car is parked 62 m from the base of the building.
What is the angle of depression from the top of the building to the car?
Give your answer to the nearest **minute**. ***(2 marks)***

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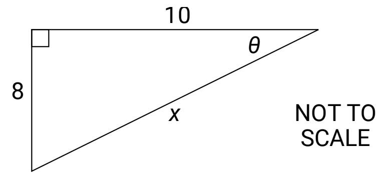
29. Measurement, STD1 M3 2020 HSC 11

Consider the triangle shown.



a. Find the value of θ, correct to the nearest degree. ***(2 marks)***

b. Find the value of x, correct to one decimal place. ***(2 marks)***

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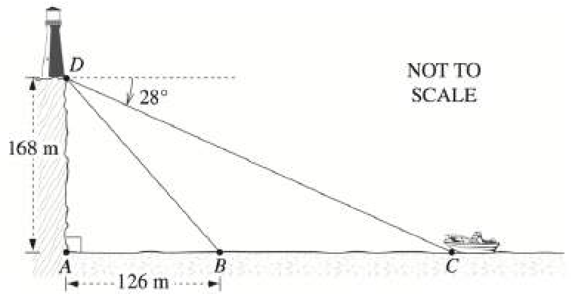
30. Measurement, STD2 M6 2010 HSC 24d

The base of a lighthouse, D, is at the top of a cliff 168 metres above sea level. The angle of depression from D to a boat at C is 28°. The boat heads towards the base of the cliff, A, and stops at B. The distance AB is 126 metres.



i. What is the angle of depression from D to B, correct to the nearest degree? ***(3 marks)***

ii. How far did the boat travel from C to B, correct to the nearest metre? ***(2 marks)***

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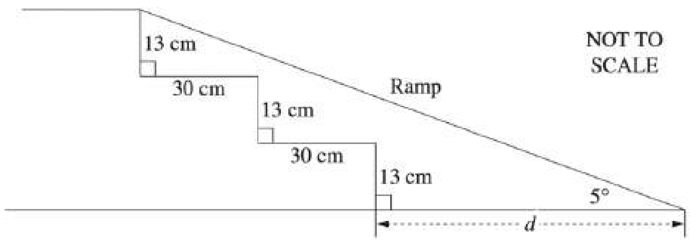
31. Measurement, STD2 M6 2012 HSC 27d

A disability ramp is to be constructed to replace steps, as shown in the diagram.

The angle of inclination for the ramp is to be 5°.

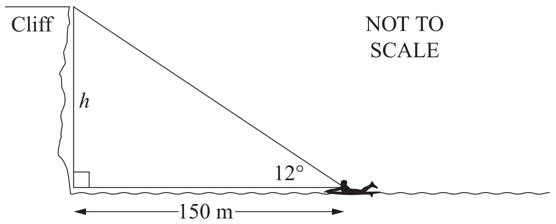


Calculate the extra distance, d, that the ramp will extend beyond the bottom step.

Give your answer to the nearest centimetre. ***(3 marks)***

32. Measurement, STD1 M3 2019 HSC 12

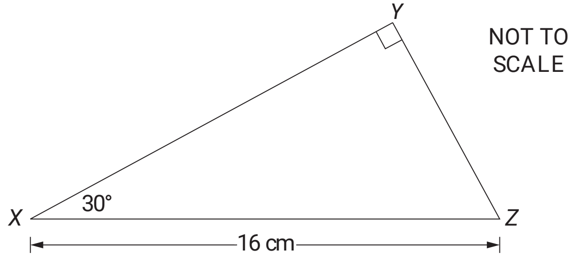
A surfer is 150 metres out to sea. From that point, the angle of elevation to the top of the cliff is 12°.



How high is the cliff, to the nearest metre? ***(2 marks)***

33. Measurement, STD1 M3 2021 HSC 28

A right-angled triangle XYZ is shown. The length of XZ is 16 cm and $\angle YXZ = 30^\circ$.

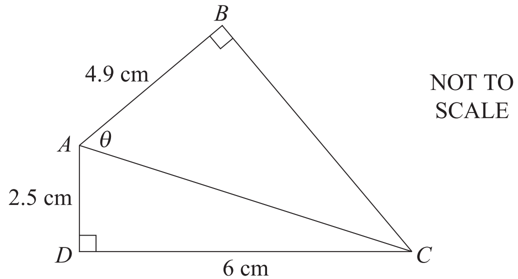


a. Find the side length, XY, of the triangle in centimetres, correct to two decimal places. ***(2 marks)***

b. Hence, find the area of triangle XYZ in square centimetres, correct to one decimal place. ***(3 marks)***

34. Measurement, STD1 M3 2019 HSC 31

Two right-angled triangles, ABC and ADC, are shown.



Calculate the size of angle θ, correct to the nearest minute. ***(3 marks)***

Worked Solutions

1. Measurement, STD2 M6 2011 HSC 9 MC

Using Pythagoras

$$\begin{aligned} c^2 &= 12^2 + 9^2 \\ &= 144 + 81 \\ &= 225 \\ \therefore c &= 15, \quad c > 0 \end{aligned}$$

$\Rightarrow C$

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2. Measurement, STD2 M6 2013 HSC 4 MC

$$\sin\theta = \frac{81}{135}$$

$\therefore \theta = 37^\circ$

$\Rightarrow B$

Note that $\tan\theta$ or $\cos\theta$ could also be used.

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3. Measurement, STD2 M6 2004 HSC 5 MC

$$\begin{aligned} \tan 20^\circ &= \frac{\text{opposite}}{\text{adjacent}} \\ &= \frac{c}{a} \end{aligned}$$

$\Rightarrow D$

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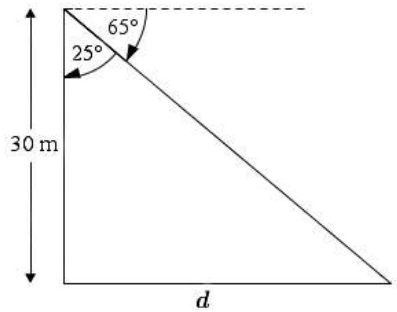
4. Measurement, STD2 M6 2006 HSC 3 MC



Let $d =$ distance from base to tree

$$\begin{aligned} \tan 25^\circ &= \frac{d}{30} \\ \therefore d &= 30 \times \tan 25^\circ \\ &= 13.98\ldots \text{ m} \end{aligned}$$

$\Rightarrow B$

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5. Measurement, STD2 M6 2007 HSC 8 MC

$$\begin{aligned} \sin 50^\circ &= \frac{12}{MN} \\ MN &= \frac{12}{\sin 50^\circ} \\ &= 15.66\ldots \end{aligned}$$

$\Rightarrow C$

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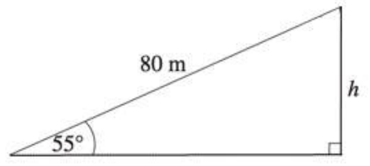
6. Measurement, STD2 M6 2008 HSC 14 MC

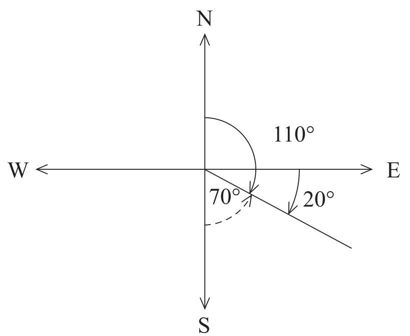


$$\begin{aligned}\sin 55^\circ &= \frac{h}{80}\\ h &= 80 \times \sin 55^\circ\\ &= 65.532\ldots \text{ m}\end{aligned}$$
$\Rightarrow B$

7. Measurement, STD2 M6 2005 HSC 8 MC

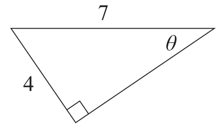
$$\begin{aligned}\tan\theta &= 85\\ \theta &= \tan^{-1} 85\\ &= 89.33^\circ\end{aligned}$$

8. Measurement, STD2 M1 2019 HSC 4 MC



$110^\circ = \text{S}70^\circ\text{E}$
$\Rightarrow C$

9. Measurement, STD2 M6 SM-Bank 1 MC



$\Rightarrow$ D

10. Measurement, STD2 M6 2012 HSC 4 MC

♦ Mean mark 42%

$$\begin{aligned}\cos 40^\circ &= \frac{x}{29}\\ x &= 29 \times \cos 40^\circ\end{aligned}$$
$\Rightarrow A$

11. Measurement, STD2 M6 2009 HSC 4 MC

$$\begin{aligned}\cos 30^\circ &= \frac{8}{x}\\ \therefore x &= \frac{8}{\cos 30^\circ}\end{aligned}$$
$\Rightarrow A$

12. Measurement, STD2 M6 2011 HSC 4 MC

$\angle$ Elevation (worm) $= 40^\circ$ (alternate angles)

$$\begin{aligned}\tan 40^\circ &= \frac{h}{15}\\ \therefore h &= 15 \times \tan 40^\circ\\ &= 12.58\ldots \text{ m}\end{aligned}$$
$\Rightarrow C$

13. Measurement, STD2 M6 2010 HSC 3 MC

Using Pythagoras:

$$\begin{aligned} GH^2 &= 12^2 + (16-11)^2 \\ &= 144 + 25 \\ &= 169 \end{aligned}$$

$$\begin{aligned} \therefore\ GH &= \sqrt{169} \\ &= 13 \text{ m} \end{aligned}$$

$\Rightarrow B$

14. Measurement, STD2 M6 2017 HSC 8 MC

$$\begin{aligned} \tan\theta &= \frac{\text{opp}}{\text{adj}} \\ &= \frac{5.3}{1.9} \\ &= 2.789\ldots \end{aligned}$$

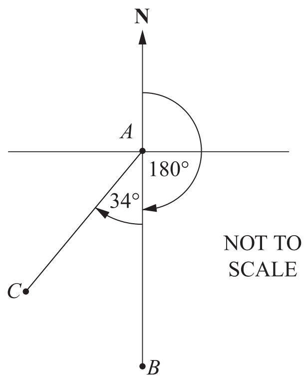
$$\begin{aligned} \therefore\ \theta &= 70.277\ldots^\circ \\ &= 70^\circ 16' 39.8'' \\ &= 70^\circ 17' \end{aligned}$$

$\Rightarrow$ B

COMMENT: An angle that has over 30" (seconds) is rounded up to the next minute (i.e. rounded up to 70°17').

15. Measurement, STD2 M6 2018 HSC 7 MC

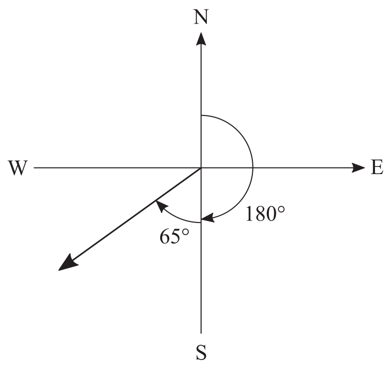
Bearing of Town C from Town A:



$$\begin{aligned} \text{Bearing} &= 180 + 34 \\ &= 214^\circ \end{aligned}$$

$\Rightarrow C$

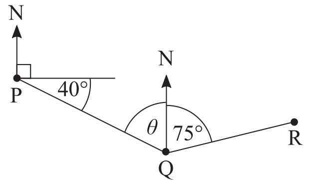
16. Measurement, STD2 M6 SM-Bank 3 MC



$$\begin{aligned} \text{True bearing} &= 180 + 65 \\ &= 245^\circ \end{aligned}$$

$\Rightarrow C$

17. Measurement, STD2 M6 SM-Bank 4 MC



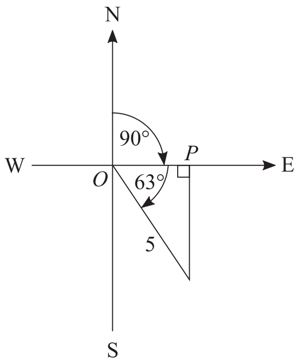
$$\begin{aligned}\theta &= 90 - 40 \quad (180^\circ \text{ in } \Delta)\\ &= 50^\circ\end{aligned}$$

$$\begin{aligned}\therefore \angle PQR &= 50 + 75\\ &= 125^\circ\end{aligned}$$

$\Rightarrow D$

---

18. Measurement, STD2 M6 SM-Bank 7 MC



Jeet finishes at P

Find OP:

$$\cos 63^\circ = \frac{OP}{5}$$

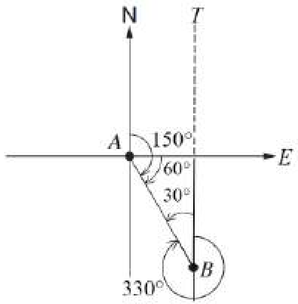
$$\begin{aligned}\therefore OP &= 5 \times \cos 63^\circ\\ &= 2.26\ldots\end{aligned}$$

$\Rightarrow A$

---

19. Measurement, STD2 M6 2010 HSC 10 MC

♦♦ *Mean mark 34%*



$\angle TBA = 30^\circ \quad$ (angle sum of triangle)

$\therefore$ Bearing of A from B

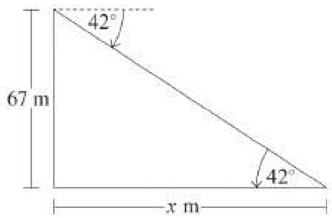
$$\begin{aligned}&= 360 - 30\\ &= 330^\circ\end{aligned}$$

$\Rightarrow D$

---

20. Measurement, STD2 M6 2015 HSC 9 MC

♦ *Mean mark 49%.*
COMMENT: The angle of depression is a regularly examined concept. Make sure you know exactly what it refers to.



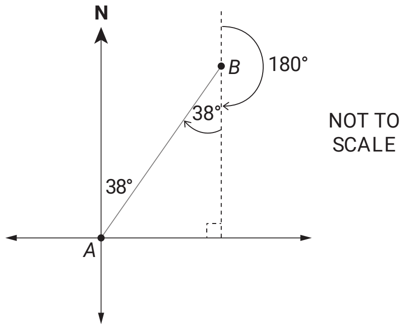
Let $x =$ distance of buoy from cliff base

$$\begin{aligned}\tan 42^\circ &= \frac{67}{x}\\ x\tan 42^\circ &= 67\\ x &= \frac{67}{\tan 42^\circ}\\ &= 74.41\ldots\text{ m}\end{aligned}$$

$\Rightarrow B$

---

21. Measurement, STD1 M3 2021 HSC 10 MC



♦♦♦ Mean mark 20%.

$$\text{Bearing } (A \text{ from } B) = 180 + 38$$
$$= 218°$$

$\Rightarrow B$

---

22. Measurement, STD2 M1 2004 HSC 23a

i. $\text{Area of } \Delta ABC = \dfrac{1}{2} \times b \times h$

$$= \frac{1}{2} \times 10 \times 5.1$$
$$= 25.5 \text{ m}^2$$

$$\text{Area of } \Delta ACD = \frac{1}{2} \times 10 \times 6.3$$
$$= 31.5 \text{ m}^2$$

$$\therefore \text{ Total Area} = 25.5 + 31.5$$
$$= 57 \text{ m}^2 \text{ ... as required}$$

ii. $V = Ah$

$$= 57 \times 0.05$$
$$= 2.85 \text{ m}^3$$

iii. $\text{Bags to buy} = \dfrac{2.85}{0.25}$

$$= 11.4$$

$\therefore$ She needs to buy 12 bags.

iv. Using Pythagoras,

$$AB^2 = 6.0^2 + 5.1^2$$
$$= 36 + 26.01$$
$$= 62.01$$
$$AB = 7.874\ldots$$
$$= 8 \text{ m (nearest metre)}$$

23. Measurement, STD2 M6 2005 HSC 25b

i. ΔABC is right-angled if $\quad a^2 + b^2 = c^2$

$$\begin{aligned} a^2+b^2 &= 5^2+12^2 \\ &= 169 \\ &= 13^2 \\ &= c^2 \dots \text{as required.} \end{aligned}$$

MARKER'S COMMENT: Know your calculator process for producing an angle in minutes/seconds. Note >30 "seconds" rounds up to the higher "minute"!

ii. $\sin \angle ABC = \dfrac{12}{13}$

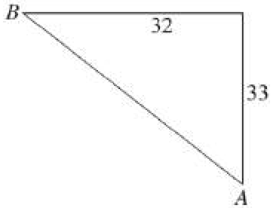
$$\begin{aligned} \therefore \angle ABC &= 67.38\dots^\circ \\ &= 67^\circ 22' 48'' \\ &= 67^\circ 23' \quad \text{(nearest minute)} \end{aligned}$$

24. Measurement, STD2 M6 2014 HSC 26b

$$\sin 28^\circ = \frac{5}{h}$$

$$\begin{aligned} \therefore h &= \frac{5}{\sin 28^\circ} \\ &= 10.6502\ldots \\ &= 10.65 \;\; \text{(2 d.p.)} \end{aligned}$$

25. Measurement, STD2 M6 2016 HSC 26d



Using Pythagoras,

$$\begin{aligned} AB &= \sqrt{32^2+33^2} \\ &= 45.967\dots \\ &= 46 \text{ m} \;\; \text{(nearest m)} \end{aligned}$$

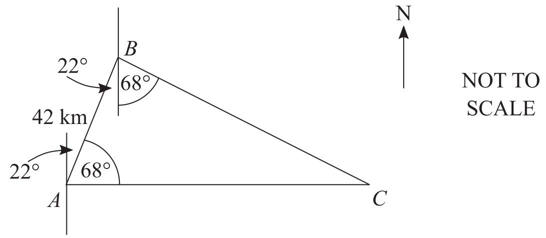
26. Measurement, STD2 M6 2017 HSC 26d

Let x = depth needed

$$\begin{aligned} \sin 2^\circ &= \frac{x}{15\,000} \\ x &= 15\,000 \times \sin 2^\circ \\ &= 523.49\dots \\ &= 523 \text{ mm} \;\; \text{(nearest mm)} \end{aligned}$$

27. Measurement, STD2 M6 SM-Bank 4

i.



$$\begin{aligned} \angle ABC &= 22+68 \\ &= 90^\circ \end{aligned}$$

ii. In ΔABC,

$$\begin{aligned} \cos\angle BAC &= \frac{AB}{AC} \\ \cos 68^\circ &= \frac{42}{AC} \\ AC &= \frac{42}{\cos 68^\circ} \\ &= 112.11\dots \\ &= 112 \text{ km} \;\; \text{(nearest km)} \end{aligned}$$

iii. Travel time $= \dfrac{\text{dist}}{\text{speed}}$

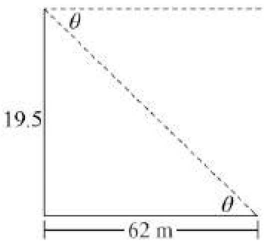
$$\begin{aligned} &= \frac{42}{12.6} \\ &= 3.333\dots \\ &= 3\text{h } 20 \text{ mins} \end{aligned}$$

28. Measurement, STD2 M6 2009 HSC 23a

i. Need to prove height (h) $\approx$ 19.5 m

$$\tan 38^\circ = \frac{h}{25}$$
$$h = 25 \times \tan 38^\circ$$
$$= 19.5321...$$
$$\approx 19.5 \text{ m} \ \ \text{... as required.}$$

ii.



Let $\angle$ Elevation (from car) $= \theta$

$$\tan\theta = \frac{h}{62}$$
$$= \frac{19.5}{62}$$
$$= 0.3145...$$
$$\therefore \theta = 17.459...$$
$$= 17^\circ 27' 33''..$$
$$= 17^\circ 28' \ \text{(nearest minute)}$$

$$\therefore \angle \text{Depression to car} = 17^\circ 28' \ \text{(alternate to } \theta)$$

♦♦ *Mean mark 33%*
MARKER'S COMMENT: If >30 "seconds", round to the higher "minute".

---

29. Measurement, STD1 M3 2020 HSC 11

a. $\tan\theta = \dfrac{8}{10}$

$$\theta = \tan^{-1}\frac{8}{10}$$
$$= 38.659...$$
$$= 39^\circ \ \text{(nearest degree)}$$

b. Using Pythagoras:

$$x = \sqrt{8^2 + 10^2}$$
$$= 12.806...$$
$$= 12.8 \ \text{(to 1 d.p.)}$$

♦ *Mean mark 44% and 39% for part (a) and (b) respectively.*

30. Measurement, STD2 M6 2010 HSC 24d

♦♦ *Mean mark 31%*

i. $$\tan\angle ADB = \frac{126}{168}$$

$$\angle ADB = 36.8698...$$

$$= 36.9^\circ \quad \text{(to 1 d.p)}$$

$$\angle\text{Depression } D \text{ to } B = 90 - 36.9$$

$$= 53.1$$

$$= 53^\circ \text{ (nearest degree)}$$

ii. Find CB:

♦♦ *Mean mark 31%*
MARKER'S COMMENT: Solve efficiently by using right-angled trigonometry. Many students used non-right angled trig, adding to the calculations and the difficulty.

$$\angle ADC + 28 = 90$$

$$\angle ADC = 62^\circ$$

$$\tan 62^\circ = \frac{AC}{168}$$

$$AC = 168 \times \tan 62^\circ$$

$$= 315.962...$$

$$CB = AC - AB$$

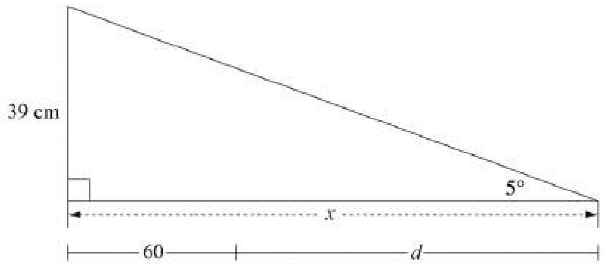
$$= 315.962... - 126$$

$$= 189.962...$$

$$= 190 \text{ m (nearest m)}$$

---

31. Measurement, STD2 M6 2012 HSC 27d



Let the horizontal part of the ramp $= x$ cm

♦♦ *Mean mark 35%*
MARKER'S COMMENT: The better responses used a diagram of a simplified version of the ramp as per the Worked Solution.

$$\tan 5^\circ = \frac{39}{x}$$

$$x = \frac{39}{\tan 5^\circ}$$

$$= 445.772...$$

$$\text{Since } x = 60 + d$$

$$d = 445.772 - 60$$

$$= 385.772 \text{ cm}$$

$$= 386 \text{ cm} \quad \text{(nearest cm)}$$

---

32. Measurement, STD1 M3 2019 HSC 12

♦♦ *Mean mark 33%.*

$$\tan 12^\circ = \frac{h}{150}$$

$$h = 150 \times \tan 12^\circ$$

$$= 31.88...$$

$$= 32 \text{ metres} \quad \text{(nearest m)}$$

---

33. Measurement, STD1 M3 2021 HSC 28

a. $\cos 30^\circ = \dfrac{XY}{16}$

$$\begin{aligned} XY &= 16 \times \cos 30^\circ \\ &= 13.856... \\ &= 13.86 \text{ cm} \ \ (2 \text{ d.p.}) \end{aligned}$$

♦♦ *Mean mark part (a) 24%.*

b. Using the sine rule:

$$\begin{aligned} \text{Area } \Delta XYZ &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} \times 16 \times 13.86 \times \sin 30^\circ \\ &= 55.44 \\ &= 55.4 \text{ cm}^2 \ \ (1 \text{ d.p.}) \end{aligned}$$

♦♦♦ *Mean mark part (b) 14%.*

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34. Measurement, STD1 M3 2019 HSC 31

Using Pythagoras in ΔACD:

$$\begin{aligned} AC^2 &= 2.5^2 + 6^2 \\ &= 42.25 \\ \therefore AC &= 6.5 \text{ cm} \end{aligned}$$

♦♦♦ *Mean mark 15%.*

In ΔABC:

$$\cos \theta = \frac{4.9}{6.5}$$

$$\begin{aligned} \theta &= \cos^{-1} \frac{4.9}{6.5} \\ &= 41.075... \\ &= 41^\circ 4' 31'' \\ &= 41^\circ 5' \ \ (\text{nearest minute}) \end{aligned}$$
